

Evolutionary coupling range varies widely among enzymes depending on selection pressure

Julian Echave^{1,*}

¹Instituto de Ciencias Físicas, Escuela de Ciencia y Tecnología, Universidad Nacional de San Martín, San Martín, Buenos Aires, Argentina

ABSTRACT Recent studies proposed that enzyme-active sites induce evolutionary constraints at long distances. The physical origin of such long-range evolutionary coupling is unknown. Here, I use a recent biophysical model of evolution to study the relationship between physical and evolutionary couplings on a diverse data set of monomeric enzymes. I show that evolutionary coupling is not universally long-range. Rather, range varies widely among enzymes, from 2 to 20 Å. Furthermore, the evolutionary coupling range of an enzyme does not inform on the underlying physical coupling, which is short range for all enzymes. Rather, evolutionary coupling range is determined by functional selection pressure.

SIGNIFICANCE Until recently, only residues near enzyme-active sites were thought to be evolutionarily constrained. However, recent studies proposed that active sites induce long-range evolutionary constraints. This seems to conflict with the common finding that physical couplings in proteins are short range. This raises the question of how short-range physical couplings may cause long-range evolutionary couplings. Here, I show that the function that maps physical coupling into evolutionary coupling depends on functional selection pressure. Under weak selection, both couplings are similarly short ranged; under strong selection, short-range physical coupling is nonlinearly turned into long-range evolutionary coupling. Thus, due to a huge variation of selection pressure, evolutionary coupling range varies widely among enzymes, from very short (2 Å) to very long (20 Å).

As enzymes evolve, different sites evolve at different rates. The main reason for such variation of evolutionary rate among sites within proteins is selection for stability (1–4). Until recently, activity constraints were thought to affect just the few residues directly involved in catalysis and their immediate neighbors (1,5). However, recent studies have reported that active sites influence evolutionary rates at long distances, slowing down the evolution of residues as distant as 30 Å (6–10). It seems reasonable to assume that such long-range evolutionary coupling could result from long-range physical couplings, such as those involved in allosteric mutations (11,12). This would align with the notion that enzymes are evolutionarily designed to optimize long-range coupling (13–15). However, except perhaps for a few allosteric residues, I would expect physical coupling to be a typical short-range, exponentially decreasing function of distance characteristic of indirect through-the-contact-network couplings (14,16–20). If this is the case, it

would leave long-range evolutionary coupling begging explanation.

The aim of this work is to verify whether physical coupling is short-range and, in that case, to study how such a short-range physical coupling may give rise to a long-range evolutionary coupling. To this end, I use the Stability-Activity Model of enzyme evolution, MSA (21). I previously showed that this model reproduces quantitatively the observed slow increase of evolutionary rate with distance from the active site that led to the proposal of long-range evolutionary couplings (7). This makes MSA suitable for exploring the physical underpinnings of long-range evolutionary coupling.

Before describing the model, I start with some definitions. The evolutionary rate, K , is the number of amino acid substitutions per unit time along an evolutionary trajectory; K_0 is the rate for the case in which all mutations are neutral; $\omega = \frac{K}{K_0}$ is the rate relative to the neutral rate; and $1 - \omega = \frac{K_0 - K}{K_0}$ is the relative slowdown with respect to the neutral evolution case. As selection pressure increases, evolution slows down: K and ω decrease, and $1 - \omega$ (the relative slowdown) increases. K , ω , and $1 - \omega$ contain exactly

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*Correspondence: jechave@unsam.edu.ar

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the same information. Therefore, in what follows, to measure the effect of selection on evolution, I will mostly use $1 - \omega$, the evolutionary slowdown.

With the help of the MSA model, I define physical and evolutionary coupling measures, and derive the formula that relates them. The MSA model is described in detail in the [Supporting material](#), Section 1 and Section 3, and in (21). MSA predicts that the evolutionary slowdown of an enzyme residue r due to selection on stability and activity is given by the following equation (Eq. S19):

$$1 - \omega(r) = 1 - \langle \min(1, e^{-a_S \Delta \Delta G(r)}) \min(1, e^{-a_A \Delta \Delta G^*(r)}) \rangle, \quad (1)$$

where $\Delta \Delta G(r)$ and $\Delta \Delta G^*(r)$ are, respectively, mutational changes of folding free energy and activation free energy; a_S and a_A are positive parameters that represent selection pressure on stability and activity, respectively; and $\langle \dots \rangle$ stands for averaging over mutations. Equation 1 relates the evolutionary slowdown of residue r to the effects of mutating this residue on stability and activity. Because here I am only interested in selection on activity, I consider a hypothetical scenario in which selection on stability is

turned off. Replacing $a_S = 0$ in Eq. 1, it follows that the evolutionary slowdown of site r due to selection on activity is given by the following equation (Eq. S20):

$$1 - \omega_A(r) = 1 - \langle \min(1, e^{-a_A \Delta \Delta G^*(r)}) \rangle. \quad (2)$$

The mutational activation free energy change $\Delta \Delta G^*(r)$ is due to the distortion of the active site caused by mutating residue r (21) (Eq. S45; [Supporting material](#), Section 1.1 and Section 3.3). Therefore, $\Delta \Delta G^*(r)$ represents the physical coupling between the enzyme's active site and residue r . This causes the slowdown $1 - \omega_A(r)$, which, therefore, can be considered a measure of evolutionary coupling. Thus, Eq. 2 governs how evolutionary coupling ($1 - \omega_A$) depends on physical coupling ($\Delta \Delta G^*$) and functional selection pressure (a_A). For notational simplicity, I will drop the explicit reference to residue r whenever possible.

I studied the relationship between physical and evolutionary couplings on a data set of monomeric enzymes used previously (21) ([Supporting material](#), Section 1.2). Briefly, for each protein, I calculated $\Delta \Delta G$ and $\Delta \Delta G^*$ for all mutations at all residues using the linearly forced elastic network model. Then, I obtained the model parameters a_S

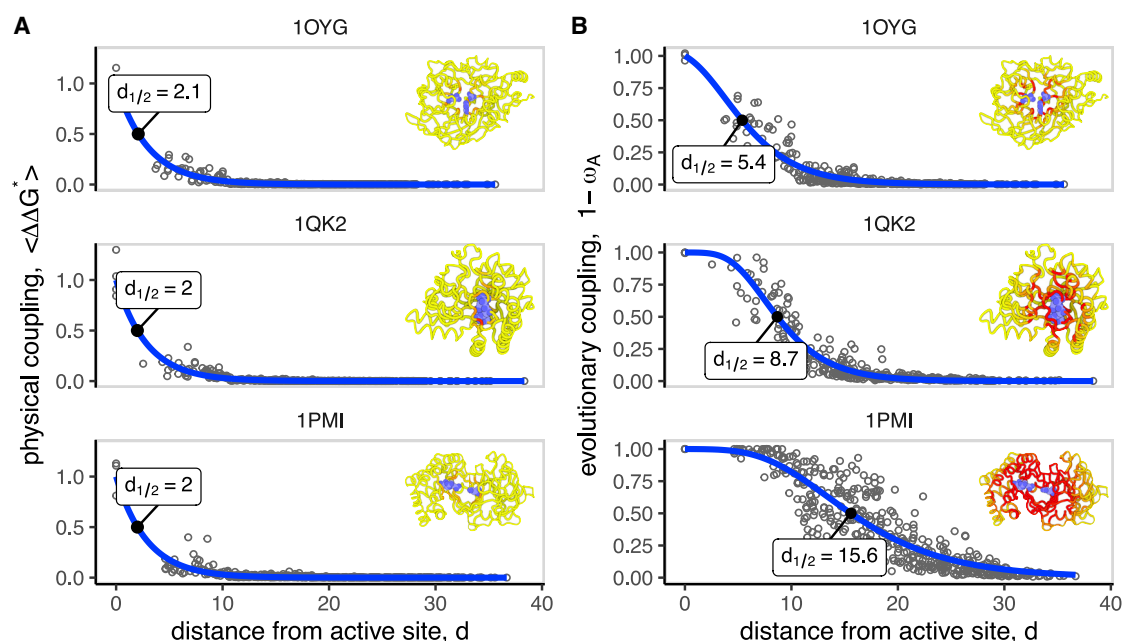


FIGURE 1 Coupling between the active site and other residues for three illustrative examples. The three cases shown are the enzymes with Protein Data Bank identity codes 1OYG, 1QK2, and 1PMI. (A) Physical coupling, as measured by the change in activation free energy that results from mutations; each point corresponds to a protein site; a site's $\langle \Delta \Delta G^* \rangle$ is the average of $\Delta \Delta G^*$ over mutations; the smooth line is an exponential fit $\langle \Delta \Delta G^* \rangle = b_0 e^{-b_1 d}$, where d is the distance from the closest active-site residue, measured in Å; the point for which coupling is half of its maximum is displayed in black; the insets show the three-dimensional (3D) protein structures colored from yellow to red according to increasing physical coupling. (B) Evolutionary coupling, as measured by $1 - \omega_A$, the relative slowdown of evolution due to selection on activity; each point corresponds to a protein site; the smooth line is a function $1 - \omega_A = 1 - e^{-c_0 e^{-c_1 d}}$, fit to the points; the point for which coupling is half of the maximum is displayed in black; the insets show the 3D protein structures colored from yellow to red according to increasing evolutionary coupling. For the sake of comparison, couplings are scaled so that the smooth fits are 1 at $d = 0$. 3D images were made with <https://3dproteinimaging.com/protein-imager> (22). To see the figure in color, go online.

and a_A by fitting MSA predictions to empirical rates. Finally, I calculated the residue-dependent evolutionary couplings, $1 - \omega_A$, and physical couplings, $\langle \Delta \Delta G^* \rangle$. For details of the calculation, see [Supporting material](#), Section 1.1. I consider MSA to be validated in a previous study (21). However, for completeness, in [Supporting material](#), Section 2, I show the excellent agreement between MSA predictions and empirical rates ([Supporting material](#), Section 2.1) and I discuss the adequacy of using the linearly forced elastic network model to calculate $\Delta \Delta G$ and $\Delta \Delta G^*$ ([Supporting material](#), Section 2.2). In what follows, I focus on the study of physical and evolutionary couplings.

For clarity, I start by considering three illustrative examples ([Fig. 1](#)). I measure coupling range using $d_{1/2}$, the distance at which coupling is half of the maximum. Physical coupling ($\langle \Delta \Delta G^* \rangle$) is similar for the three examples: it decreases exponentially with increasing distance, and it has very short range ($d_{1/2}^{\text{phys}}$ is 2.1, 2.0, and 2.0 Å for 1OYG, 1QK2, and 1PMI, respectively; [Fig. 1 A](#)). In contrast, evolutionary coupling ($1 - \omega_A$) varies among the examples, it is not an exponential but a sigmoid, and its range varies widely ($d_{1/2}^{\text{evo}}$ is 5.4, 8.7, and 15.6 Å for 1OYG, 1QK2, and 1PMI, respectively; [Fig. 1 B](#)). It will be shown below that this variation of evolutionary coupling range is due to the variation of functional selection pressure acting on these three enzymes (a_A is 29.2, 80.2, and 800 for 1OYG, 1QK2, and 1PMI, respectively).

[Fig. 2](#) shows the distance-dependence of physical and evolutionary couplings for all the enzymes studied. Physical coupling is a short-range exponential decline with distance, very similar for all cases ([Fig. 2 A](#)). In contrast, the distance-dependence of evolutionary coupling varies among proteins,

from a short-range exponential decline to a long-range sigmoidal decline ([Fig. 2 B](#)). Because physical coupling is very similar for all enzymes, from [Eq. 2](#), it follows that the variation among enzymes of evolutionary coupling must be determined by the selection parameter a_A . This is confirmed in [Fig. 2 C](#), which shows that evolutionary coupling range is independent of physical coupling range, and [Fig. 2 D](#), that shows that the variation of evolutionary coupling range is almost completely explained by the selection pressure parameter a_A .

Thus, according to [Fig. 2](#), evolutionary coupling range varies widely among enzymes, from 1.9 to 19.7 Å, as a result of the variation of parameter a_A over more than four orders of magnitude, from 6×10^{-2} to 2×10^3 . This huge variation of a_A represents a variation of the functional selection pressure under which enzymes evolve ([Supporting material](#), Section 2.3). This variation can be explained by [Eq. 2](#), that nonlinearly maps a short-range, exponentially decreasing physical coupling into a sigmoidally decreasing evolutionary coupling whose range increases with selection pressure ([Supporting material](#), Section 2.4). In summary, the fundamental finding of this work is that long-range evolutionary coupling is not due to enzymes being particularly well designed for long-range physical coupling, but it is a consequence of a nonlinear amplification of physical coupling under strong functional selection pressure.

The previous findings provide a mechanism that explains how functional constraints slow down enzyme evolution. A priori, the decrease of the rate of evolution with increasing selection pressure could be uniformly distributed among all residues. However, this work implies a different picture: increasing functional selection pressure increases the range of influence of the enzyme's active site on other

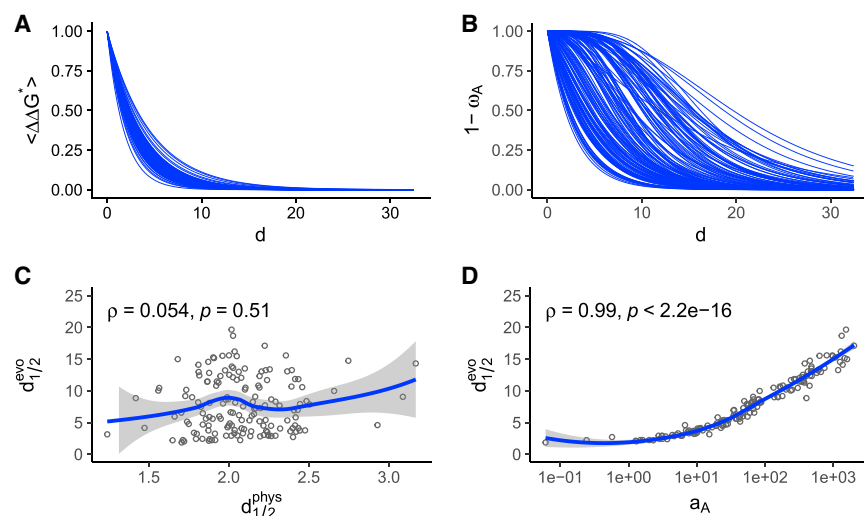


FIGURE 2 Evolutionary coupling range increases with selection pressure on activity. Coupling between the active site and other residues for each protein of a data set of 157 monomeric enzymes of diverse sizes, structures, and functions. (A) Physical coupling, as measured by $\langle \Delta \Delta G^* \rangle$, the activation free energy change averaged over mutations at the mutated site; each line is the smooth fit $\langle \Delta \Delta G^* \rangle = b_0 e^{-b_1 d}$ for one protein of the data set, where d is the distance from the closest active site residue in Å. (B) Evolutionary coupling, as measured by $1 - \omega_A$, the relative slowdown due to selection on activity; each line is the smooth fit $1 - \omega_A = 1 - e^{-c_0 e^{-c_1 d}}$ for one protein of the data set; Couplings are scaled so that all smooth fits are 1 at $d = 0$. (C) Range of evolutionary coupling measured by $d_{1/2}^{\text{evo}}$, the distance at which $1 - \omega_A$ be-

comes half of its maximum, versus the range of physical coupling, measured by $d_{1/2}^{\text{phys}}$, the distance at which $\langle \Delta \Delta G^* \rangle$ becomes half of its maximum. (D) Range of evolutionary coupling measured by $d_{1/2}^{\text{evo}}$ versus functional selection pressure measured by model parameter a_A . In (C) and (D), each point represents one protein, ρ is Spearman correlation coefficient, p is the p -value, and the lines are local regression fits. To see the figure in color, go online.

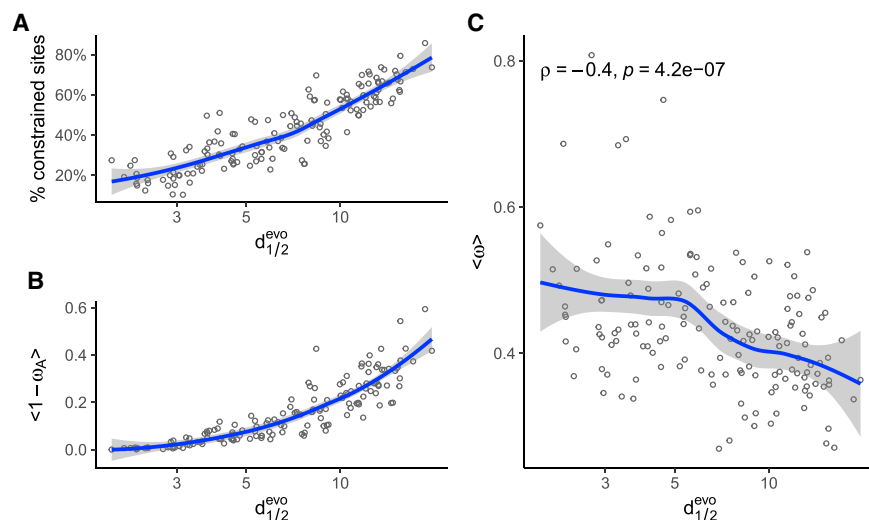


FIGURE 3 Increasing evolutionary coupling range slows down enzyme evolution. Evolutionary coupling range is measured by $d_{1/2}^{evo}$, the distance at which evolutionary coupling is half its maximum. (A) Increase of the fraction of functionally constrained sites; the number of activity-constrained residues was calculated using

$$n_A = e^{-\sum_r p_r \ln p_r}, \text{ where } r \text{ denotes residue and } p_r = (1 - \omega_A') / \sum_r (1 - \omega_A'). \text{ So defined,}$$

n_A , which varies between 1 and the total number of sites, measures how distributed over sites $1 - \omega_A$ is. (B) Increase of overall enzyme coupling, measured by $1 - \omega_A$ averaged over residues. (C) Predicted decrease of the protein rate of evolution, measured by the relative rate ω averaged over residues; ρ is Spearman's correlation coefficient, p is the p -value. In all panels, each point represents one protein and lines are local regression fits. To see the figure in color, go online.

residues (Fig. 2 D); as this range increases, more sites become functionally constrained (Fig. 3 A) and the active site becomes more tightly coupled to the rest of the protein (Fig. 3 B); as a result, the enzyme's evolution slows down (Fig. 3 C). In this work, I have derived a_A from patterns of rate variation among sites within proteins. However, the model predicts that a_A should also influence rate variation among proteins, which would connect rate variation within proteins with rate variation among proteins. Further work is needed to test this important prediction.

To finish, I mention another two research directions suggested by this work. First, for enzymes, functional selection pressure depends on metabolic role (23–26). Specifically, the main functional constraints are enzyme-specific metabolic flow and enzyme essentiality (24). Therefore, these properties should correlate with parameter a_A of this work, and, as a consequence, metabolic role should affect evolutionary coupling range. This prediction should be verified. Second, these findings indicate the intriguing possibility of manipulating evolutionary coupling range by adjusting selection pressure, which could be explored using enzyme evolution experiments.

DATA AVAILABILITY

The data and code underlying this article are available in Zenodo, at <http://doi.org/10.5281/zenodo.4309233>.

SUPPORTING MATERIAL

Supporting material can be found online at <https://doi.org/10.1016/j.bpj.2021.08.042>.

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SUPPORTING CITATIONS

References (27–38) appear in the [Supporting materials and methods](#).

REFERENCES

- Echave, J., S. J. Spielman, and C. O. Wilke. 2016. Causes of evolutionary rate variation among protein sites. *Nat. Rev. Genet.* 17:109–121.
- Echave, J., and C. O. Wilke. 2017. Biophysical models of protein evolution: understanding the patterns of evolutionary sequence divergence. *Annu. Rev. Biophys.* 46:85–103.
- Bastolla, U., Y. Dehouck, and J. Echave. 2017. What evolution tells us about protein physics, and protein physics tells us about evolution. *Curr. Opin. Struct. Biol.* 42:59–66.
- Goldstein, R. A., and D. D. Pollock. 2017. Sequence entropy of folding and the absolute rate of amino acid substitutions. *Nat. Ecol. Evol.* 1:1923–1930.
- Bartlett, G. J., C. T. Porter, ..., J. M. Thornton. 2002. Analysis of catalytic residues in enzyme active sites. *J. Mol. Biol.* 324:105–121.
- Dean, A. M., C. Neuhauser, ..., G. B. Golding. 2002. The pattern of amino acid replacements in alpha/beta-barrels. *Mol. Biol. Evol.* 19:1846–1864.
- Jack, B. R., A. G. Meyer, ..., C. O. Wilke. 2016. Functional sites induce long-range evolutionary constraints in enzymes. *PLoS Biol.* 14:e1002452.
- Sharir-Ivry, A., and Y. Xia. 2018. Nature of long-range evolutionary constraint in enzymes: insights from comparison to pseudoenzymes with similar structures. *Mol. Biol. Evol.* 35:2597–2606.

9. Sharir-Ivry, A., and Y. Xia. 2019. Using pseudoenzymes to probe evolutionary design principles of enzymes. *Evol. Bioinform. Online*. 15:1176934319855937.
10. Sharir-Ivry, A., and Y. Xia. 2019. Non-catalytic binding sites induce weaker long-range evolutionary rate gradients than catalytic sites in enzymes. *J. Mol. Biol.* 431:3860–3870.
11. Perica, T., Y. Kondo, ..., S. A. Teichmann. 2014. Evolution of oligomeric state through allosteric pathways that mimic ligand binding. *Science*. 346:1254346.
12. Guarnera, E., and I. N. Berezovsky. 2019. On the perturbation nature of allostery: sites, mutations, and signal modulation. *Curr. Opin. Struct. Biol.* 56:18–27.
13. Flechsig, H. 2017. Design of elastic networks with evolutionary optimized long-range communication as mechanical models of allosteric proteins. *Biophys. J.* 113:558–571.
14. Naganathan, A. N. 2019. Modulation of allosteric coupling by mutations: from protein dynamics and packing to altered native ensembles and function. *Curr. Opin. Struct. Biol.* 54:1–9.
15. Campitelli, P., T. Modi, ..., S. B. Ozkan. 2020. The role of conformational dynamics and allostery in modulating protein evolution. *Annu. Rev. Biophys.* 49:267–288.
16. Maslov, S., and I. Ispolatov. 2007. Propagation of large concentration changes in reversible protein-binding networks. *Proc. Natl. Acad. Sci. USA*. 104:13655–13660.
17. Maslov, S., K. Sneppen, and I. Ispolatov. 2007. Spreading out of perturbations in reversible reaction networks. *New J. Phys.* 9:273.
18. Rajasekaran, N., S. Suresh, ..., A. N. Naganathan. 2017. A general mechanism for the propagation of mutational effects in proteins. *Biochemistry*. 56:294–305.
19. Rajasekaran, N., and A. N. Naganathan. 2017. A self-consistent structural perturbation approach for determining the magnitude and extent of allosteric coupling in proteins. *Biochem. J.* 474:2379–2388.
20. Rajasekaran, N., A. Sekhar, and A. N. Naganathan. 2017. A universal pattern in the percolation and dissipation of protein structural perturbations. *J. Phys. Chem. Lett.* 8:4779–4784.
21. Echave, J. 2019. Beyond stability constraints: a biophysical model of enzyme evolution with selection on stability and activity. *Mol. Biol. Evol.* 36:613–620.
22. Tomasello, G., I. Armenia, and G. Molla. 2020. The Protein Imager: a full-featured online molecular viewer interface with server-side HQ-rendering capabilities. *Bioinformatics*. 36:2909–2911.
23. Boucher, J. I., D. N. A. Bolon, and D. S. Tawfik. 2016. Quantifying and understanding the fitness effects of protein mutations: laboratory versus nature. *Protein Sci.* 25:1219–1226.
24. Aguilar-Rodríguez, J., and A. Wagner. 2018. Metabolic determinants of enzyme evolution in a genome-scale bacterial metabolic network. *Genome Biol. Evol.* 10:3076–3088.
25. Alvarez-Ponce, D., B. Sabater-Muñoz, ..., M. A. Fares. 2016. Essentiality is a strong determinant of protein rates of evolution during mutation accumulation experiments in *Escherichia coli*. *Genome Biol. Evol.* 8:2914–2927.
26. Poyatos, J. F. 2020. Genetic buffering and potentiation in metabolism. *PLoS Comput. Biol.* 16:e1008185.
27. Echave, J. 2008. Evolutionary divergence of protein structure: the linearly forced elastic network model. *Chem. Phys. Lett.* 457:413–416.
28. Echave, J., and F. M. Fernández. 2010. A perturbative view of protein structural variation. *Proteins*. 78:173–180.
29. Ming, D., and M. E. Wall. 2005. Allostery in a coarse-grained model of protein dynamics. *Phys. Rev. Lett.* 95:198103.
30. Furnham, N., G. L. Holliday, ..., J. M. Thornton. 2014. The Catalytic Site Atlas 2.0: cataloging catalytic sites and residues identified in enzymes. *Nucleic Acids Res.* 42:D485–D489.
31. Webb, E. C. 1992. Enzyme Nomenclature. Academic Press, San Diego, CA.
32. McCandlish, D. M., and A. Stoltzfus. 2014. Modeling evolution using the probability of fixation: history and implications. *Q. Rev. Biol.* 89:225–252.
33. Echave, J., E. L. Jackson, and C. O. Wilke. 2015. Relationship between protein thermodynamic constraints and variation of evolutionary rates among sites. *Phys. Biol.* 12:025002.
34. Huang, T.-T., M. L. del Valle Marcos, ..., J. Echave. 2014. A mechanistic stress model of protein evolution accounts for site-specific evolutionary rates and their relationship with packing density and flexibility. *BMC Evol. Biol.* 14:78.
35. Marcos, M. L., and J. Echave. 2015. Too packed to change: side-chain packing and site-specific substitution rates in protein evolution. *PeerJ*. 3:e911.
36. Stein, R. L. 2011. Kinetics of Enzyme Action: Essential Principles for Drug Hunters. Wiley, Hoboken, NJ.
37. Schowen, R. L. 1978. Chapter 20. Catalytic power and transition-state stabilization. In *Transition States of Biochemical Processes* R. D. Gandour and R. L. Schowen, eds.. Springer, pp. 77–144.
38. Micheletti, C., P. Carloni, and A. Maritan. 2004. Accurate and efficient description of protein vibrational dynamics: comparing molecular dynamics and Gaussian models. *Proteins*. 55:635–645.

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Supplemental information

Evolutionary coupling range varies widely among enzymes depending on selection pressure

Julian Echave

Evolutionary coupling range varies widely among enzymes depending on selection pressure

Supporting Material

Julian Echave

Instituto de Ciencias Físicas, UNSAM and CONICET, San Martín, Buenos Aires, Argentina.

1 Supplementary Methods

1.1 Summary of the MSA model

This section summarises the stability-activity enzyme evolution model, MSA, and gathers its key equations. A more detailed description of the model and derivation of the equations is presented in Section 3 and in [1]. To avoid duplication, I use the equation numbers of Section 3, where the formulas are derived.

Rates and slowdowns, definitions The evolutionary rate, K , is the number of amino-acid substitutions per unit time along an evolutionary trajectory; K_0 is the rate for the case in which all mutations are neutral; $\omega = \frac{K}{K_0}$ is the rate relative to the neutral rate; and $1 - \omega = \frac{K_0 - K}{K_0}$ is the relative slowdown with respect to the neutral evolution case. As selection pressure increases, evolution slows down: K and ω decrease, while $1 - \omega$ increases. To measure the effect of selection on evolution, I use $1 - \omega$, the evolutionary slowdown.

MSA slowdowns MSA predicts that the relative slowdown of residue r is given by:

$$1 - \omega(r) = 1 - \langle \min(1, e^{-a_S \Delta\Delta G(r)}) \min(1, e^{-a_A \Delta\Delta G^*(r)}) \rangle \quad (\text{S19})$$

where $\Delta\Delta G(r)$ and $\Delta\Delta G^*(r)$ are, respectively, mutational changes of folding free energy and activation free energy; a_S and a_A are site-independent positive parameters that represent selection pressure on stability and activity, respectively; and $\langle \dots \rangle$ stands for averaging over mutations at site r . Eq. S19 relates the slowdown of the evolution of a site to the effects of mutating this site on stability and activity.

Physical and evolutionary couplings In a hypothetical scenario in which selection on stability is turned off, the slowdown caused by selection on activity is given by:

$$1 - \omega_A(r) = 1 - \langle \min(1, e^{-a_A \Delta\Delta G^*(r)}) \rangle \quad (\text{S20})$$

where $\Delta\Delta G^*(r)$ represents the *physical coupling* between residue r and the enzyme’s active site; $1 - \omega_A(r)$ represents the *evolutionary coupling* between residue r and the active site; Eq. S20 relates these two couplings.

LFENM calculation of $\Delta\Delta G$ To calculate $\Delta\Delta G$ and $\Delta\Delta G^*$, I used the Linearly Forced Elastic Network Model (LFENM) [2, 3]. LFENM represents a given protein as an elastic network of nodes (amino acids) connected by harmonic springs (interactions) and models mutations as random perturbations of the lengths of the springs that connect the mutated site to other sites. The elastic network has a quadratic energy function with a single minimum at the equilibrium conformation. (A protein conformation is represented by $\mathbf{r}$, the vector of Cartesian coordinates of all network nodes.) Given a wild type protein with energy $V_{\text{wt}}(\mathbf{r})$ with minimum at $\mathbf{r}_{\text{wt}}^0$ and a mutant with energy $V_{\text{mut}}(\mathbf{r})$ with minimum at $\mathbf{r}_{\text{mut}}^0$. The difference between the equilibrium conformations is given by:

$$\mathbf{r}_{\text{mut}}^0 - \mathbf{r}_{\text{wt}}^0 = -\mathbf{K}^{-1}\mathbf{f} \quad (\text{S25})$$

where $\mathbf{f}$ is a “force” vector related to the perturbations applied to model the mutation. The free-energy change caused by the mutation is:

$$\Delta\Delta G_{\gamma\gamma_0}(r) = \frac{1}{2} \sum_{j < i} k_{ij} \delta l_{ij}^2 - \frac{1}{2} (\mathbf{r}_{\text{mut}}^0 - \mathbf{r}_{\text{wt}}^0)^T \mathbf{K} (\mathbf{r}_{\text{mut}}^0 - \mathbf{r}_{\text{wt}}^0) \quad (\text{S28})$$

where k_{ij} is the force constant of the spring connecting sites i and j , δl_{ij} is the change of the length of spring ij due to the mutation from wild-type state γ_0 to the mutated state γ , and $\mathbf{K}$ is the Hessian matrix common to $V_{\text{mut}}(\mathbf{r})$ and $V_{\text{wt}}(\mathbf{r})$. $\Delta\Delta G$ given by Eq. S28 is the difference between the mutant’s and wild-type’s minimum energies (see Figure S8). Using these, Eq. S29 allows the calculation for arbitrary $\alpha \rightarrow \beta$ mutations.

LFENM calculation of $\Delta\Delta G^*$ To calculate $\Delta\Delta G^*$ I made the following considerations [1]. First, at low substrate concentrations, the activation energy barrier is the free energy difference between the transition state ES^* and enzyme E and substrate S free in solution: $\Delta G^* = G(ES^*) - G(E) - G(S)$. Second, ΔG^* can be written as the sum of two contributions, a *distortion energy*, which is the energy necessary for the enzyme and substrate to adopt their transition state conformations, E^* and S^* , plus a *vertical binding energy*, which is the energy released when E^* and S^* bind to form ES^* . Third, I assume that mutations have no effect on the vertical binding energy or the substrate distortion, but affect only the enzyme’s distortion free energy: $\Delta\Delta G^* = \Delta G(E \rightarrow E^*)$. Finally, I assume that the wild-type has a preorganized active site with a conformation which is identical to the transition state conformation. Using these assumptions

and the LFENM model, I derived that $\Delta\Delta G^*$ is the energy needed to distort the mutant's active site from the mutant's conformation $\mathbf{r}_{a,\text{mut}}^0$ to the wild type conformation $\mathbf{r}_{a,\text{wt}}^0$ (Figure S8), which is given by:

$$\Delta\Delta G_{\gamma\gamma_0}^*(r) = \frac{1}{2}(\mathbf{r}_{a,\text{wt}}^0 - \mathbf{r}_{a,\text{mut}}^0)^T \mathbf{K}_{aa}^{\text{eff}}(\mathbf{r}_{a,\text{wt}}^0 - \mathbf{r}_{a,\text{mut}}^0) \quad (\text{S45})$$

where $\mathbf{K}_{aa}^{\text{eff}}$ is a matrix that allows the calculation of the effective energy of distortions within the conformational subspace spanned by active residue coordinates $\mathbf{r}_a$. Eq. S45 corresponds to mutations starting at wild-type state γ_0 ; From these, Eq. S46 allows the calculation of other $\alpha \rightarrow \beta$ mutations.

1.2 Model parameters

Elastic Network Model There is a large variety of ENMs. Here, I used the ENM of Ming and Wall [4]: amino acids are represented by single nodes; nodes are connected if they are within $R_0 = 10.5\text{\AA}$; l_{ij} is the distance between the nodes in the pdb structure; the force constants are $k_{ij} = 189\text{ Kcal/Mol}$ for sequence neighbors and $k_{ij} = 4.5\text{ Kcal/Mol}$ otherwise. I placed nodes at the side-chain geometric centers (except for glycine, which has no side chain, for which I used C_α coordinates).

LFENM perturbations A mutation is emulated by perturbing independently each of the springs connected to the mutated site by adding perturbations δl_{ij} . Here, I used $\delta l_{ij} \sim N(0, \sigma = 0.3\text{\AA})$.

Calculation of model parameters a_S and a_A

LFENM mutational scan To calculate model rates for a given protein, I started by performing a full mutational scan. For each enzyme, I used its pdb structure to build the ENM. Then, at each site I introduced $N = 19$ LFENM mutations and, for each mutation, I calculated $\Delta\Delta G$ (Eq. S28) and $\Delta\Delta G^*$ (Eq. S45).

Fitting of parameters a_S and a_A Then, I calculated model rates $K(r)$ using Eq. S13. The model parameters a_S and a_A were obtained by minimizing the Residual Sum of Squares $RSS = \sum_r (K_{\text{obs}}(r) - K(r))^2$ using the general purpose optimization function `optim` of R package `stats`.

Calculation of $\langle \Delta\Delta G^* \rangle$ and $1 - \omega_A$ Finally, I averaged over mutations at each site to obtain the site-specific physical coupling, $\langle \Delta\Delta G^* \rangle$, and evolutionary coupling $1 - \omega_A$.

1.3 Data set

I studied data set of 157 enzymes, a subset of the 524 enzymes used in [5]. Specifically, I kept only monomeric enzymes and removed those that had missing amino acids or broken chains, which could result in wrong elastic network models. I also discarded 3 proteins for which the non-linear fitting of the model parameters did not converge. Catalytic residue information was obtained from the Catalytic Site Atlas [6]. The structures of these proteins were obtained from the RCSB protein database[7]. The data set is diverse: no two enzymes have more than 25% sequence identity and there are representatives of the main SCOP structural classes [8] and of the six main EC functional classes [9]. For some details of the data set, see Figure S1

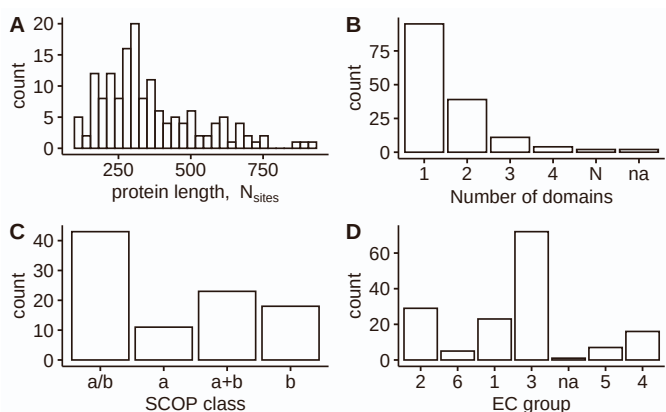


Figure S1: **Properties of the enzymes studied.** The data set is composed of 157 monomeric enzymes. **A:** distribution of sequence lengths. **B:** distribution of number of domains. **C:** distribution of SCOP class for mono-domain cases. **D:** distribution of EC group, the first classification level of the Enzyme Commission number.

1.4 Empirical rates

I used site-specific “observed” rates K_{obs} calculated before by Jack et al [5]. Briefly, estimating K_{obs} involves finding homologous sequences, aligning them, inferring a phylogeny, and using the sequence alignment and phylogeny as input for the program `rate4site` to estimate the substitution rate of each site [10].

2 Supplementary Results

2.1 MSA assessment

The model MSA predicts site-specific evolutionary rates from mutational changes of free energy ($\Delta\Delta G$), and activation free energy ($\Delta\Delta G^*$). MSA was extensively assessed against other models and empirical data in [1] where it was shown to fit empirical rates and their variation with distance from the active site. For completeness, a brief assessment is replicated here. Fig. S2 shows that the correlation coefficient between predicted and observed rates varies in the range $0.19 \leq R \leq 0.80$, with an average of $R = 0.62$, which demonstrates the excellent agreement between predicted and empirical rates.

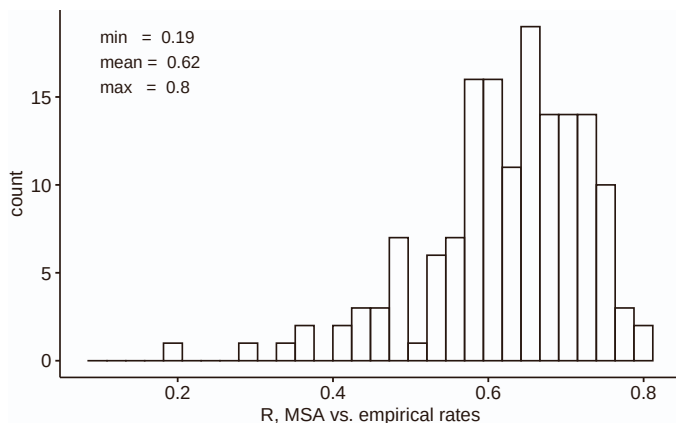


Figure S2: **MSA rates vs. empirical rates.** Distribution of the correlation coefficient R between rates predicted by the model MSA and empirical rates, inferred from multiple sequence alignments using the program Rate4Site by Jack et al. [5].

2.2 On the use of LFENM to calculate $\Delta\Delta G$ and $\Delta\Delta G^*$

In this section, I consider three possible caveats related to using the Linearly Forced Elastic network model (LFENM) to calculate $\Delta\Delta G$ and $\Delta\Delta G^*$. First, *a priori*, a possible limitation of LFENM is that being a coarse grained model, it might not predict $\Delta\Delta G$ and $\Delta\Delta G^*$ accurately. However, In previous studies I have shown that predictions performed using a stability-based model, MS, with LFENM $\Delta\Delta G$ gives very similar results to the presumably better all-atom method FoldX, that has been especially parametrised to calculate $\Delta\Delta G$ values. This is confirmed by Fig. S3, that shows that there is no significant difference between the fit to empirical of rates obtained with LFENM or FoldX. If anything LFENM gives slightly better results.

Second, a related issue is whether LFENM is able to predict $\Delta\Delta G^*$. At this

point, I am unable to fully demonstrate it can. However, the MSA model, which uses LFENM-calculated $\Delta\Delta G$ and $\Delta\Delta G^*$, clearly outperforms stability-based models that use either LFENM $\Delta\Delta G$ or FoldX $\Delta\Delta G$ (Fig. S3). In addition, stability-only models *fail* to reproduce the observed dependence of evolutionary rate with increasing distance from the active site, while the MSA model, by including LFENM $\Delta\Delta G^*$, succeeds in predicting the rate-distance dependence quantitatively [1]. Therefore, LFENM $\Delta\Delta G^*$ values seem to capture how coupling varies with distance, which is the interest of the present paper.

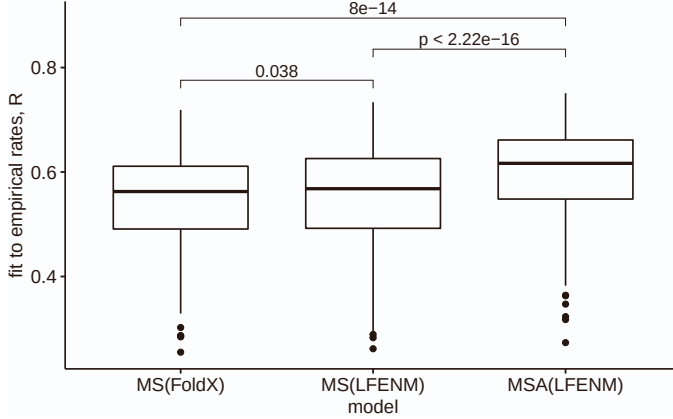


Figure S3: **Comparison of three models: MSA(LFENM), MS(LFENM), and MS(FoldX).** MSA(LFENM) is model MSA that uses $\Delta\Delta G$ and $\Delta\Delta G^*$ calculated using the LFENM coarse-grained model; MS(LFENM) is a stability-only model that uses LFENM $\Delta\Delta G$ values; MS(FoldX) is the same stability model but based on FoldX $\Delta\Delta G$ values. R is the correlation coefficient between model predictions and empirical rates. Means are compared Wilcoxon’s signed-rank test for paired cases.

A final limitation to discuss is that LFENM models random perturbations, not actual amino-acid mutations. In elastic network models, sequence is not explicit, but implicit in the protein’s conformation. A wild-type node state γ_0 is implicit in $\mathbf{r}_{\text{wt}}^0$, and a mutant state γ is implicit in $\mathbf{r}_{\text{mut}}^0$. Since mutants are obtained using random perturbations, these node states cannot be mapped to actual natural amino acids. Therefore, $\Delta\Delta G$ and $\Delta\Delta G^*$ calculated using LFENM do not correspond to actual amino-acid mutations. All the same, LFENM $\Delta\Delta G$ and $\Delta\Delta G^*$ do get the statistics correctly. MSA rates, which are averages over perturbations, are in very good agreement with observed rates, which are averages over actual amino-acid mutations [1]. Thus, LFENM is suitable for the study of the variation of rates among sites.

2.3 On the meaning of parameter a_A and its huge variation

As shown in Figure 2 of the main text, evolutionary coupling range varies widely among enzymes, from 1.9 Å to 19.7 Å, as a result of a huge variation of parameter a_A over 4 orders of magnitude, from 6×10^{-2} to 2×10^3 . Why does a_A vary so widely? The MSA model assumes that the only relevant enzyme property is $\Delta\Delta G^*$, and that a_A represents external selection pressure. However, there might be some enzyme property missed by $\Delta\Delta G^*$ that influences a_A . In other words, a_A could be affected by internal or external factors. As possible internal factors, I considered several properties related to protein size, shape, or architecture. I found that a_A is independent of protein length, average solvent accessibility, average packing density, number of domains, and SCOP structural class (see Figure S4). This independence from internal properties suggests that the huge variation of a_A among enzymes probably reflects a variation of *external* factors affecting enzyme evolution; a_A would represent external selection pressure on catalytic activity. Therefore, I take a_A to stand for external functional selection pressure. It's large variation among enzymes would be consistent with the finding that functional constraints related to enzyme metabolic roles vary very widely [11].

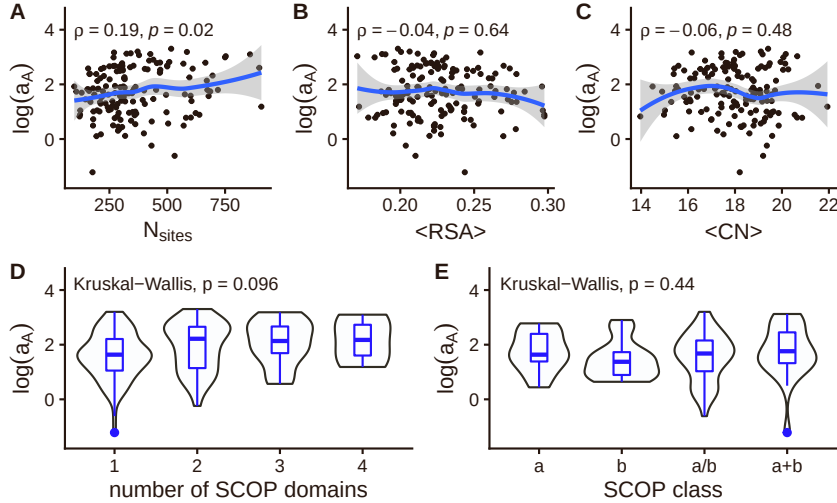


Figure S4: **Non-dependence of a_A on internal enzyme properties.** Model parameter a_A vs. various internal properties: **A:** protein length, N_{sites} ; **B:** average Relative Solvent Accessibility ($\langle RSA \rangle$), which increases with protein size, depending on shape; **C:** average Contact Number ($\langle CN \rangle$), a property of the contact network that decreases with size; **D:** number of SCOP domains; **E:** SCOP structural class. ρ is Spearman's correlation coefficient; p stands for p-value; the Kruskal-Wallis test was used to compare means.

2.4 Evolutionary coupling as non-linearly amplified physical coupling

As described in the main text, I found that while physical coupling displays a very similar short-range exponential decrease with distance from the active residues, the distance-dependence of evolutionary coupling varies widely among enzymes, depending on functional selection pressure. This is due to the non-linear dependence of evolutionary coupling on physical coupling (see Eq. S20 and Figure S5). For very small a_A , Eq. S20 can be expanded to first order, which leads to: $1 - \omega_A \approx a_A < \Delta\Delta G^* >$, so that at very low selection pressures both couplings are short-range exponential declines. For larger a_A , the short-range exponential physical coupling is non-linearly mapped into a long-range sigmoidal evolutionary coupling. Therefore, long-range evolutionary coupling is due to the non-linear amplification of physical coupling under strong functional selection pressure.

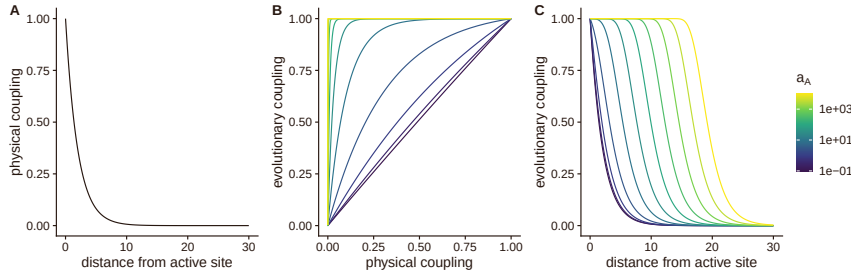


Figure S5: Selection pressure controls how physical coupling is mapped into evolutionary coupling. **A:** typical exponentially decreasing physical coupling. **B:** evolutionary coupling vs. physical coupling for increasing functional selection pressure (Eq. S20: $1 - \omega_A = 1 - < \min(1, e^{-a_A \Delta\Delta G^*}) >$). **C:** distance-dependence of evolutionary coupling, which is the composition of panels **A** and **B**. Note that the same short-range physical coupling (panel **A**) is mapped into evolutionary couplings with increasingly longer range as functional selection pressure increases (panel **C**). The curves of panels **B** and **C** are coloured according the functional selection pressure parameter a_A .

3 Theory

This section describes the Stability-Activity Model of enzyme evolution, MSA. The present material is adapted from the article where the model was presented, where the reader is referred to for a more detailed description, assessment and discussion [1]. The section is structured as follows. First, I derive the model. Specifically, I pose the fitness landscape (Figure S6A); derive the fixation probability (Figure S6B); derive a mean-field approximation to this fixation probability (Figure S6C); derive the formulas needed to calculate site-specific amino-acid substitution rates (Eq. S13); and derive the function that relates evolutionary coupling to physical coupling (Eq. S20.) Second, I derive a formula to calculate stability changes $\Delta\Delta G$ using the "Linearly-Forced Elastic Network Model" (LFENM) (Eq. S28 and Figure S6D). Finally, I derive a formula to calculate mutational activation free energy changes $\Delta\Delta G^*$ using LFENM (Eq. S45 and Figure S6D).

3.1 The MSA model

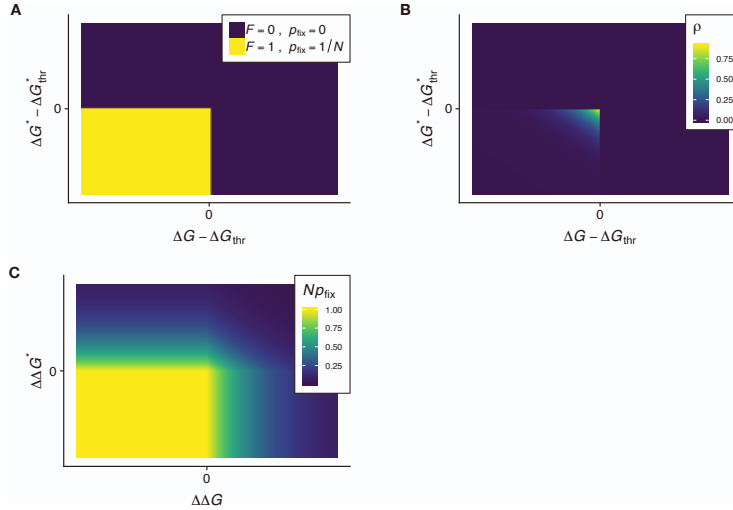


Figure S6: **The stability-activity model MSA.** **A:** Fitness is a 2D step function: enzymes are viable ($F = 1$) and may become fixed ($p^{fix} = 1/N$, N being population size) only if their folding free energy ΔG is more negative than a threshold energy ΔG_{thr} and if their activation energy barrier ΔG^* is lower than a threshold ΔG_{thr}^* . **B:** Distribution of ΔG and ΔG^* at evolutionary equilibrium. **C:** A mean field p^{fix} that depends only on $\Delta\Delta G$ (mutational change of stability) and $\Delta\Delta G^*$ (mutational change of activation energy) is obtained by integrating the p^{fix} of panel A over the distribution of panel B, . This figure is part of a figure used before in [1].

Fitness landscape

The stability-activity model of enzyme evolution MSA is specified by a fitness function that depends on enzyme stability, quantified by the folding free energy ΔG , and activity, quantified by the activation energy of the enzymatic reaction ΔG^* . Specifically, fitness is given by the following two-dimensional step function (Figure S6A):

$$F(\Delta G, \Delta G^*) = \begin{cases} 1 & \text{if } \Delta G < \Delta G_{\text{thr}} \text{ and } \Delta G^* < \Delta G_{\text{thr}}^* \\ 0 & \text{if } \Delta G \geq \Delta G_{\text{thr}} \text{ or } \Delta G^* \geq \Delta G_{\text{thr}}^* \end{cases} \quad (\text{S1})$$

Thus, all genotypes with stability above a stability threshold ($\Delta G < \Delta G_{\text{thr}}$) and activation energy barrier below an activation energy threshold ($\Delta G^* < \Delta G_{\text{thr}}^*$) are equally viable ($F = 1$) and genotypes that are too unstable ($\Delta G^* \geq \Delta G_{\text{thr}}^*$) and/or too inactive ($\Delta G \geq \Delta G_{\text{thr}}$) are inviable ($F = 0$).

Fixation probability

Assuming that the mutation rate μ is so small that the time between mutations is much larger than the time needed to fix or discard a new mutation, evolution is an origination-fixation process in which a new mutant is either lost or fixed before a another mutant originates [12]. Consider a monoclonal population of N identical haploid genotypes with stability ΔG and activation energy ΔG^* . For the fitness function of Eq. S1, when a mutant originates, the probability that it becomes fixed, replacing all the ancestor genotypes, is given by:

$$p^{\text{fix}} = \begin{cases} \frac{1}{N} & \text{for } \Delta G + \Delta\Delta G < \Delta G_{\text{thr}} \text{ and } \Delta G^* + \Delta\Delta G^* < \Delta G_{\text{thr}}^* \\ 0 & \text{if } \Delta G + \Delta\Delta G \geq \Delta G_{\text{thr}} \text{ or } \Delta G^* + \Delta\Delta G^* \geq \Delta G_{\text{thr}}^* \end{cases} \quad (\text{S2})$$

where $\Delta G + \Delta\Delta G = \Delta G_{\text{mut}}$ is the mutant's stability and $\Delta G^* + \Delta\Delta G^* = \Delta G_{\text{mut}}^*$ is its activation free energy. If the mutant is within the allowed stability and activity thresholds, the mutation is neutral, the mutant has the same fitness as the other $N - 1$ individuals of the population, thus it will become fixed with probability $p^{\text{fix}} = 1/N$. On the other hand, if the mutant is not stable or active enough, it will necessarily become lost, thus $p^{\text{fix}} = 0$.

Mean-field fixation probability

It is usually easier to calculate relative free energy changes, such as $\Delta\Delta G$ and $\Delta\Delta G^*$, than absolute values such as ΔG and ΔG^* . Therefore, for tractability, I follow [13, 14] and derive a *mean field* approximation to the fixation probability that depends only on $\Delta\Delta G$ and $\Delta\Delta G^*$. Specifically, I assume that $\Delta\Delta G$ and $\Delta\Delta G^*$ are independent of the specific sequence background in which the

mutation happens. Then, I average the fixation probability over ΔG and ΔG^* :

$$p^{\text{fix}}(\Delta\Delta G, \Delta\Delta G^*) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p^{\text{fix}}(\Delta G_{\text{wt}}, \Delta G_{\text{wt}}^*, \Delta\Delta G, \Delta\Delta G^*) \rho_S(\Delta G_{\text{wt}}) \rho_A(\Delta G_{\text{wt}}^*) d\Delta G_{\text{wt}} d\Delta G_{\text{wt}}^* \quad (\text{S3})$$

where ρ_S is the stationary distribution of stabilities and ρ_A the distribution of activation energies. These distributions may be approximated using truncated exponentials (Figure S6B). For evolution on a stability-dependent step-function fitness function, in the rare mutation limit ($\mu N \ll 1$), the stationary stability distribution is well approximated by a truncated exponential[13]:

$$\rho_S(\Delta G) = \begin{cases} a_S e^{a_S(\Delta G - \Delta G_{\text{thr}})} & \text{if } \Delta G < \Delta G_{\text{thr}} \\ 0 & \text{if } \Delta G \geq \Delta G_{\text{thr}} \end{cases} \quad (\text{S4})$$

Since evolution in the ΔG^* dimension is mathematically equivalent, I further assume:

$$\rho_A(\Delta G^*) = \begin{cases} a_A e^{a_A(\Delta G^* - \Delta G_{\text{thr}}^*)} & \text{if } \Delta G^* < \Delta G_{\text{thr}}^* \\ 0 & \text{if } \Delta G^* \geq \Delta G_{\text{thr}}^* \end{cases} \quad (\text{S5})$$

Replacing Eq. S2, Eq. S4, and Eq. S5 into Eq. S3 and integrating, it follows that

$$p^{\text{fix}}(\Delta\Delta G, \Delta\Delta G^*) = 1/N \min(1, e^{-a_S \Delta\Delta G}) \min(1, e^{-a_A \Delta\Delta G^*}) \quad (\text{S6})$$

This is mean field approximation to the fixation probability in terms of mutational changes of stability, $\Delta\Delta G$, and activation energy, $\Delta\Delta G^*$ (see Figure S6C).

Substitution rates

The variation of evolutionary constraints within proteins is commonly quantified using site-specific substitution rates. In this section, I derive the formulas needed to calculate site-specific amino-acid substitution rates. First, let us derive site-specific substitution rates as a function of stability and activity changes. The rate of $\alpha \rightarrow \beta$ substitutions (the number of substitutions per unit time) at site r is given by [15]:

$$Q_{\beta\alpha}(r) = N\mu p_{\beta\alpha}^{\text{fix}}(r) \quad (\text{S7})$$

where N is the population size, μ is the mutation rate, and $p_{\beta\alpha}^{\text{fix}}(r)$ is the probability of fixing the $\alpha \rightarrow \beta$ mutation at site r , given by Eq. S6:

$$p_{\beta\alpha}^{\text{fix}}(r) = \frac{1}{N} \min(1, e^{-a_S \Delta\Delta G_{\beta\alpha}(r)}) \min(1, e^{-a_A \Delta\Delta G_{\beta\alpha}^*(r)}) \quad (\text{S8})$$

where $\Delta\Delta G_{\beta\alpha}(r)$ is the free energy change due to the $\alpha \rightarrow \beta$ mutation at site r , and $\Delta\Delta G_{\beta\alpha}^*(r)$ the corresponding activation energy change. Replacing Eq. S8 into Eq. S7, it follows:

$$Q_{\beta\alpha}(r) = \mu \min(1, e^{-a_S \Delta\Delta G_{\beta\alpha}(r)}) \min(1, e^{-a_A \Delta\Delta G_{\beta\alpha}^*(r)}) \quad (\text{S9})$$

Second, I derive the site-specific amino-acid probability distributions at evolutionary equilibrium. Let $\pi_\gamma(r)$ be the probability of finding amino acid γ at site r at a certain evolutionary time t . These probabilities satisfy the evolution equations:

$$\frac{d\pi_\beta(r)}{dt} = \sum_\alpha Q_{\beta\alpha}(r)\pi_\alpha(r). \quad (\text{S10})$$

At equilibrium, $d\pi_\gamma(r)/dt = 0$. From this condition and Eq. S9, it can be shown that the equilibrium probabilities are given by:

$$\pi_\alpha^{eq}(r) = \frac{e^{-a_S \Delta\Delta G_{\alpha\gamma_0}(r) - a_A \Delta\Delta G_{\alpha\gamma_0}^*(r)}}{\sum_\gamma e^{-a_S \Delta\Delta G_{\gamma\gamma_0}(r) - a_A \Delta\Delta G_{\gamma\gamma_0}^*(r)}} \quad (\text{S11})$$

where $\pi_\alpha^{eq}(r)$ is the probability of finding amino acid α at site r at evolutionary equilibrium, γ_0 stands for a reference state (amino acid) that may be chosen arbitrarily without loss of generality, and the denominator ensures that $\sum_\alpha \pi_\alpha^{eq}(r) = 1$.

Finally, I derive the total site-specific substitution rates. At evolutionary equilibrium, the total substitution rate of site r is obtained by summing over final states and averaging over initial states, which gives:

$$K(r) = \sum_\alpha \sum_{\beta \neq \alpha} Q_{\beta\alpha}(r)\pi_\alpha^{eq}(r) \quad (\text{S12})$$

where $Q_{\beta\alpha}(r)$ is given by Eq. S9 and $\pi_\alpha^{eq}(r)$ is given by Eq. S11. Therefore, replacing Eq. S9 and Eq. S11 into Eq. S12, it follows that the rate of substitutions at a site r is given by:

$$\boxed{\begin{aligned} K(r) &= \sum_\alpha \sum_{\beta \neq \alpha} Q_{\beta\alpha}(r)\pi_\alpha^{eq}(r) & (\text{S13a}) \\ Q_{\beta\alpha}(r) &= \mu \min(1, e^{-a_S \Delta\Delta G_{\beta\alpha}(r)}) \min(1, e^{-a_A \Delta\Delta G_{\beta\alpha}^*(r)}) & (\text{S13b}) \\ \pi_\alpha^{eq}(r) &= \frac{e^{-a_S \Delta\Delta G_{\alpha\gamma_0}(r) - a_A \Delta\Delta G_{\alpha\gamma_0}^*(r)}}{\sum_\gamma e^{-a_S \Delta\Delta G_{\gamma\gamma_0}(r) - a_A \Delta\Delta G_{\gamma\gamma_0}^*(r)}} & (\text{S13c}) \end{aligned}}$$

Using this equation, given selection parameters a_S and a_A , site-specific rates may be calculated from mutational changes of stability, $\Delta\Delta G_{\beta\alpha}$, and activation free energy, $\Delta\Delta G_{\beta\alpha}^*$.

Relative rate

Let K be the rate of evolution of a protein or a protein site and K_0 be the evolutionary rate in the neutral evolution case (i.e. mutations are neutral, or, equivalently, when natural selection does not operate). Then, the relative rate of evolution is defined by:

$$\omega \equiv \frac{K}{K_0} \quad (\text{S14})$$

The value of ω is related natural selection. $\omega > 1$ for positive selection (adaptive evolution), and $\omega < 1$ for negative/purifying selection.

For the MSA model, the relative slowdown is obtained by replacing Eq. S13 into definition Eq. S14 (note that $K_0 = K(a_S = 0, a_A = 0)$), which leads to:

$$\omega(r) = \frac{1}{N_{aa} - 1} \sum_{\alpha} \sum_{\beta \neq \alpha} \min(1, e^{-a_S \Delta \Delta G_{\beta\alpha}(r)}) \min(1, e^{-a_A \Delta \Delta G_{\beta\alpha}^*(r)}) \pi_{\alpha}(r)^{eq} \quad (\text{S15})$$

where $\pi_{\alpha}(r)$ is the probability of finding amino-acid α in residue r at equilibrium and N_{aa} is the number of possible amino acids (commonly 20).

The notation can be simplified by noting that for any property X that depends on mutations $\alpha \rightarrow \beta$, the average over an evolutionary trajectory is given by:

$$\langle X \rangle = \frac{1}{N_{aa} - 1} \sum_{\alpha} \sum_{\beta \neq \alpha} X_{\beta\alpha} \pi_{\alpha}^{eq} \quad (\text{S16})$$

Therefore, using this notation in Eq. S15, gives the more compact:

$$\omega(r) = \langle \min(1, e^{-a_S \Delta \Delta G(r)}) \min(1, e^{-a_A \Delta \Delta G^*(r)}) \rangle \quad (\text{S17})$$

This equation relates mutational changes $\Delta \Delta G$ and $\Delta \Delta G^*$ to relative rates of evolution ω for the model MSA. That MSA is a model of evolution under negative selection can be seen in Figure S7. ω is maximum at $a_S = a_A = 0$, when selection does not operate, and it decreases when the selection-pressure parameters a_S and/or a_A increase.

Relative slowdown

To measure the effects of selection pressure, rather than ω , it is more convenient to use $1 - \omega$, which is the *relative slowdown*:

$$1 - \omega = \frac{K_0 - K}{K_0} \quad (\text{S18})$$

For instance, a protein, or a protein site, with $\omega = 0.7$ is evolving at 70% of its unselected rate; its slowdown is $1 - \omega = 0.3$, which means that it is evolving 30% more slowly than it would be under no selection. Replacing Eq. S17 into Eq. S18, it follows that:

$$1 - \omega(r) = 1 - \langle \min(1, e^{-a_S \Delta \Delta G(r)}) \min(1, e^{-a_A \Delta \Delta G^*(r)}) \rangle \quad (\text{S19})$$

This expresses the slowdown of the evolution of a protein site as a function of the mutational changes of stability ($\Delta \Delta G$) and activation free energy ($\Delta \Delta G^*$), depending on parameters a_S , which measures selection pressure on stability, and a_A , which measures selection pressure on activity.

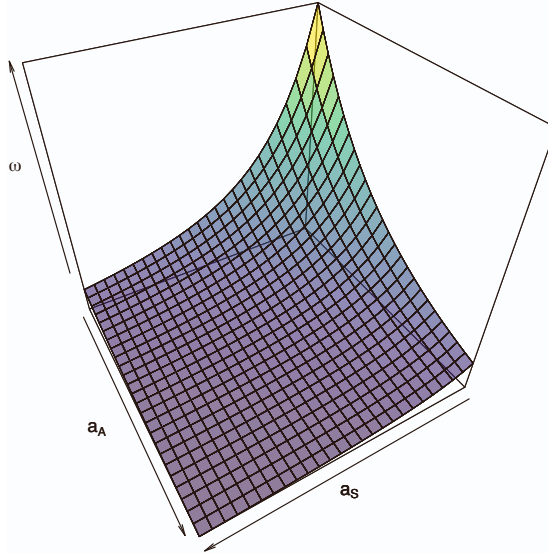


Figure S7: **Variation of the relative rate ω with selection pressure.** Plot of the relative rate ω predicted by MSA for one protein site of the protein with pdb code 1PMI, as a function of the model's selection parameters a_S and a_A . When $a_S = a_A = 0$, there is no selection and the rate is maximum. As selection on stability (a_S) or on activity (a_A) increase, ω decreases.

Slowdown due selection on activity

Eq. S19 relates the slowdown of the evolution of a site to the effects of mutating this site on stability and activity. To study the effects of selection on activity, I consider a hypothetical scenario in which selection on stability is turned off. Replacing $a_S = 0$ into Eq. S19 it follows that:

$$1 - \omega_A(r) = 1 - \langle \min(1, e^{-a_A \Delta \Delta G^*(r)}) \rangle \quad (\text{S20})$$

where $1 - \omega_A(r)$ is the slowdown of site r due to selection on activity. Note that $\omega_A = \omega(a_S = 0, a_A)$, so that it is the curve at the intersection of the $\omega(a_S, a_A)$ surface and the $a_A - \omega$ plane (see Figure S7).

Physical and evolutionary couplings

I will show below that $\Delta \Delta G^*(r)$ results from modifications at the active-site location due to mutations at residue r [1]. Therefore, $\Delta \Delta G^*(r)$ represents the *physical coupling* between residue r and the enzyme's active site. Accordingly, this physical coupling causes a slowdown $1 - \omega_A(r)$, which, therefore, represents the *evolutionary coupling* between residue r and the active site. Thus,

Eq. S20 governs how evolutionary coupling ($1 - \omega_A$) depends on physical coupling ($\Delta\Delta G^*$) and functional selection pressure (a_A).

3.2 LFENM calculation of $\Delta\Delta G$

The site-specific rates of evolution that MSA aims to predict depend on mutational changes of stability, $\Delta\Delta G$, and of activation energy, $\Delta\Delta G^*$. These are represented in Figure S8. To calculate these mutational changes, I use the Linearly Forced Elastic Network Model (LFENM) [2, 3, 16, 17]. This section describes the LFENM model and its use to calculate $\Delta\Delta G$.

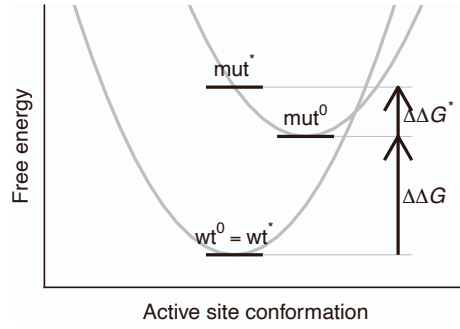


Figure S8: **Mutational impact on stability and activity.** $\Delta\Delta G$ is the difference between energy minima; $\Delta\Delta G^*$ is the energy needed to distort the mutant from its equilibrium conformation mut^0 to the active conformation mut^* (the wild type's active site is assumed preorganized in the active conformation: $\text{wt}^0 = \text{wt}^*$).

Wild type elastic network

The LFENM model represents proteins as networks of nodes, representing amino acids, connected by harmonic springs, representing interactions. The energy function of a given wild type protein is given by

$$V_{\text{wt}} = \frac{1}{2} \sum_{i < j} k_{ij} (d_{ij} - l_{ij})^2 \quad (\text{S21})$$

where d_{ij} is the distance between nodes i and j in an arbitrary conformation, and k_{ij} and l_{ij} are, respectively, the force constant and equilibrium length of the spring connecting nodes i and j . V_{wt} is usually approximated using a second-order Taylor expansion of Eq. S21:

$$V_{\text{wt}}(\mathbf{r}) = \frac{1}{2} (\mathbf{r} - \mathbf{r}_{\text{wt}}^0)^T \mathbf{K} (\mathbf{r} - \mathbf{r}_{\text{wt}}^0) \quad (\text{S22})$$

where T stands for ‘transpose’, $\mathbf{r}$ is the column vector of node coordinates that represents an arbitrary conformation, $\mathbf{r}_{\text{wt}}^0$ is the wild type’s equilibrium conformation, and $\mathbf{K}$ is the Hessian matrix of second derivatives of V_{wt} .

Mutated elastic network

The LFENM model simulates mutations by perturbing the equilibrium length of the springs that connect the mutated site to its neighbors. Thus, introducing a mutation at site r results in a mutant with energy

$$V_{\text{mut}} = \frac{1}{2} \sum_{i < j} k_{ij} (d_{ij} - l_{ij} - \delta l_{ij})^2 \quad (\text{S23})$$

where δl_{ij} is a random perturbation added to l_{ij} . (Only contacts of site r are perturbed: $\delta l_{ij} \neq 0$ only for $i = r$ or $j = r$, and $\delta l_{ij} = 0$ otherwise.)

In general, the mutation will shift the equilibrium conformation. The mutant’s equilibrium structure is given by [2]:

$$\mathbf{r}_{\text{mut}}^0 = \mathbf{r}_{\text{wt}}^0 + \delta \mathbf{r}^0, \quad (\text{S24})$$

where the mutational change of the equilibrium structure is given by

$$\delta \mathbf{r}^0 = \mathbf{r}_{\text{mut}}^0 - \mathbf{r}_{\text{wt}}^0 = -\mathbf{K}^{-1} \mathbf{f}, \quad (\text{S25})$$

where $\mathbf{f}$ is a ‘‘force’’ vector that depends on the perturbations δl_{ij} . (perturbing the spring-length parameter l_{ij} by adding δl_{ij} is equivalent to applying a force of magnitude f_{ij} in the direction of the vector that goes from site i to site j .)

Expanding the mutant’s energy up to second order around its equilibrium conformation $\mathbf{r}_{\text{mut}}^0$ leads to

$$V_{\text{mut}}(\mathbf{r}) = V_{\text{mut}}^0 + \frac{1}{2} (\mathbf{r} - \mathbf{r}_{\text{mut}}^0)^T \mathbf{K} (\mathbf{r} - \mathbf{r}_{\text{mut}}^0) \quad (\text{S26})$$

where

$$V_{\text{mut}}^0 = \frac{1}{2} \sum_{j \sim i} k_{ij} \delta l_{ij}^2 - \frac{1}{2} (\delta \mathbf{r}^0)^T \mathbf{K} \delta \mathbf{r}^0 \quad (\text{S27})$$

is the mutant’s minimum energy.

Note that $\mathbf{K}$ in the previous equations is the Hessian matrix of V_{wt} and V_{mut} ; since $\mathbf{K}$ is not affected by *linear* perturbations, $\mathbf{K}_{\text{mut}} = \mathbf{K}_{\text{wt}} = \mathbf{K}$. For more details on the LFENM see [2, 3].

LFENM formula for $\Delta \Delta G$

Consider the free energy change due to introducing a single mutation in the wild-type protein. Let γ_0 be the state of node r of the elastic network that represents the wild-type protein; and let γ represent the mutated state of node r of the elastic network representing the mutant protein, obtained through perturbing the network, as explained above. The free-energy change due to the mutation,

$\Delta\Delta G_{\gamma\gamma_0}(r)$, is the free energy difference between the mutated elastic network and the wild type elastic network (Figure S8). Using basic statistical physics, from Eq. S22, Eq. S26, and Eq. S27, it can be found that:

$$\Delta\Delta G_{\gamma\gamma_0}(r) = \frac{1}{2} \sum_{j < i} k_{ij} \delta l_{ij}^2 - \frac{1}{2} (\mathbf{r}_{\text{mut}}^0 - \mathbf{r}_{\text{wt}}^0)^T \mathbf{K} (\mathbf{r}_{\text{mut}}^0 - \mathbf{r}_{\text{wt}}^0) \quad (\text{S28})$$

where $\mathbf{r}_{\text{mut}}^0 - \mathbf{r}_{\text{wt}}^0$ may be obtained using Eq. S25 and δl_{ij} are the perturbations added to the springs of the mutated site r to model the mutation $\gamma_0 \rightarrow \gamma$. The free-energy change for an arbitrary mutation $\alpha \rightarrow \beta$ is given by:

$$\Delta\Delta G_{\beta\alpha}(r) = \Delta\Delta G_{\beta\gamma_0}(r) - \Delta\Delta G_{\alpha\gamma_0}(r) \quad (\text{S29})$$

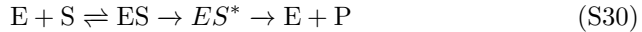
This follows from basic Thermodynamics and the fact that a mutation $\alpha \rightarrow \beta$ can be obtained via the path $\alpha \rightarrow \gamma_0 \rightarrow \beta$.

N.B: mutations of the wild-type protein are always destabilising ($\Delta\Delta G_{\gamma\gamma_0}(r) \geq 0$.) By construction, all springs of the wild-type elastic network are relaxed. As a result of the mutation, the springs connected to the mutated site become stressed, so that in the wild-type conformation $\mathbf{r}_{\text{wt}}^0$ these springs are either too compressed or too extended. This local stress is represented by the first term of Eq. S28, which is always positive. Part of this stress is relaxed in the mutant conformation $\mathbf{r}_{\text{mut}}^0$, which is represented by the negative second term of Eq. S28. However, while the local stress near the mutation is now somewhat reduced, other springs move out of their ideal equilibrium lengths, so that even in the mutant's equilibrium conformation all springs are stressed. Therefore, $\Delta\Delta G_{\gamma,\gamma_0}(r) \geq 0$ for all r and γ . In contrast, an arbitrary $\alpha \rightarrow \beta$ mutation may be destabilizing or stabilizing depending on the relative magnitudes of the two terms of Eq. S29 [1].

3.3 LFENM calculation of $\Delta\Delta G^*$

Activation energy barrier ΔG^*

Consider the following simple mechanism of enzyme catalysis:



The first reaction is the binding between enzyme and substrate to form the complex ES . The second reaction is the formation of products via a transition state ES^* . According to transition state theory, for low substrate concentrations this reaction has second order kinetics with rate constant [18]

$$k = \frac{k_B T}{h} e^{-\frac{1}{k_B T} \Delta G^*} \quad (\text{S31})$$

where k_B is Boltzmann's constant, T is the absolute temperature, h is Planck's constant, and

$$\Delta G^* = G(\text{ES}^*) - G(\text{E}) - G(\text{S}) \quad (\text{S32})$$

is the *activation free energy*, which is the free-energy difference between the enzyme-substrate activated complex and enzyme and substrate in their free solution states.

Following [19, 18], we can break the activation process into two steps, a *distortion step*, in which enzyme and substrate adopt the conformation they have in the activated complex, followed by a *vertical binding step*, in which enzyme and substrate in the right *activated poses* bind to each other without further conformational change. Then:

$$\Delta G^* = \Delta G(E \rightarrow E^*) + \Delta G(S \rightarrow S^*) + \Delta G(E^* + S^* \rightarrow ES^*) \quad (\text{S33})$$

The first term is the distortion energy needed for the enzyme to adopt its activated-complex pose, the second is the substrate distortion energy, and the third term is the vertical binding energy, the free energy released when enzyme and substrate in the right poses bind together.

$\Delta\Delta G^*$ is the mutant's distortion free energy

This section introduces two reasonable assumptions: (1) that mutations change only the enzyme distortion contribution to $\Delta\Delta G^*$, and (2) that the wild-type's active site is pre-organized into the activated pose. With this assumptions, it follows that $\Delta\Delta G^*$ is the mutant's distortion free energy. (Figure S6D.)

Assumption: mutations change only the enzyme distortion contribution. From Eq. S33, the free-energy change due to a mutation can be partitioned as:

$$\Delta\Delta G^* = \Delta\Delta G(E \rightarrow E^*) + \Delta\Delta G(S \rightarrow S^*) + \Delta\Delta G(E^* + S^* \rightarrow ES^*) \quad (\text{S34})$$

The mutation will not affect the distortion energy of the substrate in solution, therefore the second term is zero. The third term should be affected only by mutations of the residues directly involved in binding (i.e. active-site residues and substrate-binding residues). Since here I am interested in exploring the effect of mutating sites distant from the active residues, to explore long-range coupling, the third term will also likely be zero. Therefore, in this work, I assume that mutations affect only the distortion energy of the enzyme, so that only the first term of Eq. S34 contributes to $\Delta\Delta G^*$:

$$\Delta\Delta G^* = \Delta\Delta G(E \rightarrow E^*) = \Delta G_{\text{mut}}(E \rightarrow E^*) - \Delta G_{\text{wt}}(E \rightarrow E^*) \quad (\text{S35})$$

Assumption: the wild-type's active site is perfectly pre-organised. In general, enzyme active sites are pre-organized: their native conformation is very similar to that observed in the activated complex. Therefore, I further assume that the wild-type's active site is perfectly pre-organized, so that

$$\Delta G_{\text{wt}}(E \rightarrow E^*) = 0 \quad (\text{S36})$$

Result: $\Delta\Delta G^*$ is the mutant's distortion free energy. Therefore, replacing Eq. S36 into Eq. S35, it follows:

$$\boxed{\Delta\Delta G^* = \Delta G_{\text{mut}}(E \rightarrow E^*)} \quad (\text{S37})$$

Thus, the mutational activation energy change is identical to the free energy needed to distort the mutant from its equilibrium conformation to the activated pose.

$\Delta\Delta G^*$ from conformational distributions

$\Delta\Delta G^*$ can be obtained from the mutant's conformational distribution. Let $\mathbf{r}_a$ be the column vector of coordinates of the catalytic residues, and $\mathbf{r}_n$ be the vector of coordinates of the rest of the residues. The enzyme distortion free energy is the energy necessary to make the active residues adopt a specific active pose $\mathbf{r}_a^*$, regardless of the conformation of the rest of the sites $\mathbf{r}_n$. Further, since we have assumed that the wild-type protein has a perfectly pre-organised active site, the active conformation is the wild-type's conformation: $\mathbf{r}_a^* = \mathbf{r}_{a,\text{wt}}^0$. Then, from basic statistical physics it follows that the mutant's distortion free energy can be obtained from

$$\Delta G_{\text{mut}}(E \rightarrow E^*) = -k_B T \log \frac{\rho_{\text{mut}}(\mathbf{r}_{a,\text{wt}}^0)}{\rho_{\text{mut}}(\mathbf{r}_{a,\text{mut}}^0)} \quad (\text{S38})$$

where $\rho_{\text{mut}}(\mathbf{r}_a)$ is the marginal distribution of $\mathbf{r}_a$ conformations, $\mathbf{r}_{a,\text{wt}}^0$ is wild-type native conformation of the active residues (which, due to pre-organisation, is the enzyme's active pose,) and $\mathbf{r}_{a,\text{mut}}^0$ is the active site mutant's equilibrium conformation. Replacing Eq. S38 into Eq. S37, gives:

$$\Delta\Delta G^* = -k_B T \log \frac{\rho_{\text{mut}}(\mathbf{r}_{a,\text{wt}}^0)}{\rho_{\text{mut}}(\mathbf{r}_{a,\text{mut}}^0)} \quad (\text{S39})$$

This equation allows the calculation of $\Delta\Delta G^*$ given $\rho_{\text{mut}}(\mathbf{r}_a^*)$.

LFENM mutant's conformational distribution

From the previous equation, to finish the derivation of $\Delta\Delta G^*$, we need the mutant's marginal distribution function $\rho_{\text{mut}}(\mathbf{r}_a)$. We can obtain this by integration of the LFENM distributions.

The mutant's conformational distribution is given by the Boltzmann density function $\rho_{\text{mut}}(\mathbf{r}) = e^{-\beta V_{\text{mut}}(\mathbf{r})}/Z$ where the partition function Z is a normalization constant. From the LFENM's V_{mut} given by Eq. S26, it follows that:

$$\rho_{\text{mut}}(\mathbf{r}) = \frac{e^{-1/2(\mathbf{r}-\mathbf{r}_{\text{mut}}^0)^T \boldsymbol{\Sigma}^{-1}(\mathbf{r}-\mathbf{r}_{\text{mut}}^0)}}{\sqrt{2\pi|\boldsymbol{\Sigma}|}} \quad (\text{S40})$$

where $\boldsymbol{\Sigma}$ is the variance-covariance matrix, given by:

$$\boldsymbol{\Sigma} = \frac{1}{k_B T} \mathbf{K}^{-1} \quad (\text{S41})$$

(Note: $\mathbf{K}$ is singular, thus the pseudo-inverse is used.)

The marginal distribution $\rho_{\text{mut}}(\mathbf{r}_a)$ is obtained by integrating Eq. S40 over the coordinates of non-active residues $\mathbf{r}_n$, which gives:

$$\rho_{\text{mut}}(\mathbf{r}_a) = \frac{e^{-1/2(\mathbf{r}_a - \mathbf{r}_{a,\text{mut}}^0)^T \boldsymbol{\Sigma}_{aa}^{-1} (\mathbf{r}_a - \mathbf{r}_{a,\text{mut}}^0)}}{\sqrt{|2\pi \boldsymbol{\Sigma}_{aa}|}} \quad (\text{S42})$$

where $\boldsymbol{\Sigma}_{aa}$ is the block of $\boldsymbol{\Sigma}$ that corresponds to the coordinates of the active-site residues. This Marginal distribution may be thought of as the distribution of of a reduced-dimensionality energy function:

$$V_{\text{mut}}(\mathbf{r}_a) = V_{\text{mut}}(\mathbf{r}_{\text{mut}}^0) + \frac{1}{2}(\mathbf{r}_a - \mathbf{r}_{a,\text{mut}}^0)^T \mathbf{K}_{aa}^{\text{eff}} (\mathbf{r}_a - \mathbf{r}_{a,\text{mut}}^0) \quad (\text{S43})$$

where

$$\mathbf{K}_{aa}^{\text{eff}} \equiv k_B T \boldsymbol{\Sigma}_{aa}^{-1} \quad (\text{S44})$$

This matrix may be interpreted as an effective energy matrix for motion of the active residues that takes into account their direct interactions and indirect interactions via their coupling with non-active residues (see e.g. [20]).

LFENM formula to calculate $\Delta\Delta G^*$

Finally, using the LFENM mutant's marginal distribution derived above, we obtain $\Delta\Delta G^*$. Replacing Eq. S42 into into Eq. S39 and using Eq. S44, leads to:

$$\Delta\Delta G_{\gamma\gamma_0}^*(r) = \frac{1}{2}(\mathbf{r}_{a,\text{wt}}^0 - \mathbf{r}_{a,\text{mut}}^0)^T \mathbf{K}_{aa}^{\text{eff}} (\mathbf{r}_{a,\text{wt}}^0 - \mathbf{r}_{a,\text{mut}}^0) \quad (\text{S45})$$

This is the activation free energy change for introducing a mutation $\gamma_0 \rightarrow \gamma$ into site r of the wild-type protein. For arbitrary mutations $\alpha \rightarrow \beta$, $\Delta\Delta G^*$ is given by:

$$\Delta\Delta G_{\beta\alpha}^*(r) = \Delta\Delta G_{\beta\gamma_0}^*(r) - \Delta\Delta G_{\alpha\gamma_0}^*(r) \quad (\text{S46})$$

which follows from the fact that mutation $\alpha \rightarrow \beta$ can be obtained by the following sequence: $\alpha \rightarrow \gamma_0 \rightarrow \beta$.

Note that according to Eq. S45, $\Delta\Delta G_{\gamma\gamma_0}^*(r) \geq 0$, so that mutations of the wild-type always increase the activation energy barrier. The reason for this is that the wild type enzyme is assumed to be perfectly pre-organised, having zero distortion contribution to the activation energy barrier. Any mutation will shift the mutant's equilibrium structure away from the ideal active pose, increasing the deformation contribution to the free energy barrier. However, for other mutations $\Delta\Delta G_{\beta\alpha}^*(r)$, given by Eq. S46, may be positive or negative.

Supporting References

- [1] Echave, J., 2019. Beyond Stability Constraints: A Biophysical Model of Enzyme Evolution with Selection on Stability and Activity. *Mol. Biol. Evol.* 36:613–620.
- [2] Echave, J., 2008. Evolutionary Divergence of Protein Structure: The Linearly Forced Elastic Network Model. *Chem. Phys. Lett.* 457:413–416.
- [3] Echave, J., and F. M. Fernández, 2010. A perturbative view of protein structural variation. *Proteins Struct. Funct. Bioinforma.* 78:173–180.
- [4] Ming, D., and M. E. Wall, 2005. Allostery in a coarse-grained model of protein dynamics. *Phys. Rev. Lett.* 95:198103.
- [5] Jack, B. R., A. G. Meyer, J. Echave, and C. O. Wilke, 2016. Functional Sites Induce Long-Range Evolutionary Constraints in Enzymes. *PLOS Biol.* 14:e1002452.
- [6] Furnham, N., G. L. Holliday, T. A. P. de Beer, J. O. B. Jacobsen, W. R. Pearson, and J. M. Thornton, 2014. The Catalytic Site Atlas 2.0: cataloging catalytic sites and residues identified in enzymes. *Nucleic Acids Res.* D485–D489.
- [7] Berman, H. M., J. Westbrook, Z. Feng, G. Gilliland, T. N. Bhat, H. Weissig, I. N. Shindyalov, and P. E. Bourne, 2000. The Protein Data Bank. *Nucleic Acids Res.* 28:235–242.
- [8] Murzin, A. G., S. E. Brenner, T. Hubbard, and C. Chothia, 1995. SCOP: a structural classification of proteins database for the investigation of sequences and structures. *J. Mol. Biol.* 247:536–540.
- [9] Webb, E. C., 1992. Enzyme Nomenclature. Wiley, San Diego.
- [10] Mayrose, I., D. Graur, N. Ben-Tal, and T. Pupko, 2004. Comparison of Site-Specific Rate-Inference Methods for Protein Sequences: Empirical Bayesian Methods Are Superior. *Mol. Biol. Evol.* 21:1781–1791.
- [11] Aguilar-Rodriguez, J., and A. Wagner, 2018. Metabolic determinants of enzyme evolution in a genome-scale bacterial metabolic network. *Genome Biol. Evol.* 10:3076–3088.
- [12] McCandlish, D. M., and A. Stoltzfus, 2014. Modeling evolution using the probability of fixation: history and implications. *Q. Rev. Biol.* 89:225–252.
- [13] Bloom, J. D., and M. J. Glassman, 2009. Inferring Stabilizing Mutations from Protein Phylogenies: Application to Influenza Hemagglutinin. *PLoS Comput. Biol.* 5:e1000349.

- [14] Echave, J., E. L. E. Jackson, and C. O. C. Wilke, 2015. Relationship between Protein Thermodynamic Constraints and Variation of Evolutionary Rates among Sites. *Phys. Biol.* 12:025002.
- [15] Echave, J., and C. Wilke, 2017. Biophysical Models of Protein Evolution: Understanding the Patterns of Evolutionary Sequence Divergence. *Annu. Rev. Biophys.* 46:85–103.
- [16] Huang, T.-T., M. L. Del Valle Marcos, J.-K. Hwang, and J. Echave, 2014. A Mechanistic Stress Model of Protein Evolution Accounts for Site-Specific Evolutionary Rates and Their Relationship with Packing Density and Flexibility. *BMC Evol. Biol.* 14:78.
- [17] Marcos, M. L., and J. Echave, 2015. Too Packed to Change: Side-Chain Packing and Site-Specific Substitution Rates in Protein Evolution. *PeerJ* 3:e911.
- [18] Stein, R. L., 2011. Kinetics of Enzyme Action: Essential Principles for Drug Hunters. Wiley, Hoboken, New Jersey.
- [19] Schowen, R. L., 1978. Catalytic power and Transition-State stabilization. In R. D. Gandour, and R. L. Schowen, editors, Transition States of Biochemical Processes, Springer, Boston, chapter 20, 77–144.
- [20] Micheletti, C., P. Carloni, and A. Maritan, 2004. Accurate and efficient description of protein vibrational dynamics: comparing molecular dynamics and Gaussian models. *Proteins* 55:635–45.